Green Earth Wire?

Main Power Input

J1 200241-1112 Mega-Fit

V_batt+

D1 SMAJ51CA

GND

50A Main Power Relay

F1 3557-10 40A

K1 J115F11CH5VDCS61.5U Coil 5V Coil 300mA

V_batt+ PROT

Cap Discharge

R40 180 — Must be high wattage res

D4 SMD26PL-TP

5V Relay PWR

Main Relay EN

IC1 SI2318CDS-T1-GE3

R6 10k

V_batt+ PROT SW

D10 LTST-C150KRKT

R31 10k

Large DC-DC Converters (With Heatspreader)

F6 0678L9250-02

PS1 I7C4W008A120V-0C3-R

CPS_IN_PROT

COMP PWR EN

C1 180uF 63V

C2 180uF 63V

C3 180uF 63V

C5 10uF 63V

C7 10uF 63V

IC4 SI2318CDS-T1-GE3

R9 10k

R1 4.64k

R3 1k

C9 100uF 63V

C11 100uF 63V

C15 100uF 63V

C17 0.1uF 63V

TP4 TestPoint

+15V

COMP PWR CM

C13 0.1uF 63V

Computer Power Supply

F7 0678L9250-02

PS2 I7C4W008A120V-0C3-R

PSPS_IN_PROT

SWITCH PWR EN

C4 180uF 63V

C25 180uF 63V

C26 180uF 63V

C6 10uF 63V

C8 10uF 63V

IC5 SI2318CDS-T1-GE3

R10 10k

R2 28

R4 1k

C10 100uF 63V

C12 100uF 63V

C16 100uF 63V

C18 0.1uF 63V

TP5 TestPoint

+48V

POE SW PWR CM

C14 0.1uF 63V

POE Switch Power Supply

Power Outputs

F4 3557-10 7.5A Blade Fuses

COMP PWR OUT PROT

J14 43650-0228 Micro-Fit

J4 PJ-063BH

D13 LTST-C150KRKT

R34 10k

F5 3557-10 5A

SWITCH PWR OUT PROT

D14

J3 43650-0415 Micro-Fit

R35 10k

F9 3557-10 TBD

48V Aux PWR OUT PROT

48V Aux PWR Out

KeySwitch Input

J10 43650-0228 Micro-Fit

TP1 KeySwitch IN

R5 28

D7 SMD26PL-TP

D2 SMAJ51CA

C19 0.1uF

Wakeup Bootstrap Ckt

Switched PWR IN PROT

R11 10k

D8 SMD26PL-TP

R12 10k

C20 10uF 63V

C21 10uF 63V

D18 KEY SW IN

R41 1k

5V LDO Regulator

U4 MCP1793T-5002H/OT 5V 100mA

C45 1uF

C44 0.1uF

C50 0.1uF

D12 SMD26PL-TP

Q5 SQJ182EP-T1_GE3

R50 0

5V BOOTSTRAP

U2 IRS25752L

MOSFET Driver

Buck Regulator 5.3V 600mA

U10 LV2862YDDC

C41 10uF 63V

R47 100k

C49 0.1uF

C43 0.1uF

L2 SRR1208-470YL 2A 47uH

D11 SK220A 200V 2A

R48 59k

R49 10k

C42 10uF 63V

D20 PPS1545-3G 5V Logic PWR

D21 PPS1545-3G 5V Relay PWR

Vout = 0.765V (1+ (R1 / R2))

--Change log from rev 0.3--
Removed light bank relay A (F2, K2, D5, IC2, R7, D11, R32)
Removed light bank relay B (F3, K3, D6, IC3, R8, D12, R33)
Removed connector J2, J8, J9
Changed J10 to a 2 pin Molex Micro-Fit
Changed J3 to a 4 pin Molex Micro-Fit
Changed J13 to a 4 pin Molex Micro-Fit
Remove External relay circuit (IC6)
Added external status light 3 colors circuit
Changed U4 from R1560S501B-E2-FE to MCP1793T-5002H/OT because no stock
Added gate driver circuit to replace sticking relay
Changed LT1910ES8 to IRS20752L because of no stock
Added LED for keyswitch
Removed J12 Pico breakout
Removed J6 Jetson breakout

Jetson Header

J7 XG4A-4035

JET_P1 JET_P2 JET_P3 JET_P4 JET_P5 JET_P7 Jet_UART1_Tx Jet_UART1_Rx JET_P11 JET_P12 PICO OUT BIT 1 PICO OUT BIT 0 PICO IN BIT 0 PICO OUT BIT 2 JET_P17 JET_P19 JET_P21 PICO IN BIT 2 JET_P23 JET_P24 JET_P26 JET_P27 JET_P28 JET_P29 JET_P31 PICO IN BIT 1 JET_P33 JET_P35 JET_P36 JET_P37 JET_P38 JET_P39 JET_P40

Auxiallary Jetson Header

J11 PPPC202LFBN-RC

JET_P1 JET_P2 JET_P3 JET_P4 JET_P5 JET_P7 Jet_UART1_Tx Jet_UART1_Rx JET_P11 JET_P12 PICO OUT BIT 1 PICO OUT BIT 0 PICO IN BIT 0 PICO OUT BIT 2 JET_P17 JET_P19 JET_P21 PICO IN BIT 2 JET_P23 JET_P24 JET_P26 JET_P27 JET_P28 JET_P29 JET_P31 PICO IN BIT 1 JET_P33 JET_P35 JET_P36 JET_P37 JET_P38 JET_P39 JET_P40

Rpi Pico

U1 SC0915

D9 SMD26PL-TP 5V Logic PWR

C30 180uF 63V

C46 10uF 63V

C47 10uF 63V

PICO 3V3

SW1 SW_Push Reset Button

SW2 SW_Push Aux SW

Aux_SW

SWITCH PWR EN
COMP PWR EN
Main Relay EN
LED_YELLOW
LED_GREEN
PICO OUT BIT 1
PICO OUT BIT 2
PICO IN BIT 0
PICO IN BIT 1
PICO IN BIT 2
LED_RED
JET_ON
Pico UART0 Tx
Pico UART0 Rx
MUX_S0
MUX_S1
MUX_S2
Analog Mux OUT
Pico LED A
POE SW PWR CM
COMP PWR CM

JP3 Jmp_2_Bridged Jet_UART1_Rx

JP4 Jmp_2_Bridged Jet_UART1_Tx

Added new functions
GP14 LED_RED
GP3 LED_YELLOW
GP4 LED_GREEN

Pico LED B
Pico LED A

D15 D16 D17

R36 1k R37 1k R38 1k

Temp Sense ICs

PICO 3V3

U6 MCP9700T-ETT

U7 MCP9700T-ETT

C39 0.1uF

C40 0.1uF

TEMP_SENSE_1
TEMP_SENSE_2

TP9 TestPoint

TP11 TestPoint

ADC MUX

PICO 3V3

C38 0.1uF

U5 74HC4051

TP14 TestPoint

KeySW_Vin MES_ADC
COMP_Vout MES_ADC
POE_SW_Vout MES_ADC
TEMP_SENSE_1
TEMP_SENSE_2

JP7 JP5 JP6

Analog Mux OUT
MUX_S0
MUX_S1
MUX_S2

TP7 TestPoint

ADC Voltage Dividers

KeySwitch IN — R21 220k — R22 10k — TP6 TestPoint

SWITCH PWR OUT PROT — R23 220k — R24 10k — TP13 TestPoint

COMP PWR OUT PROT — R27 220k — R28 10k — TP8 TestPoint

External Status Light Power Supply 24V

+48V

U8 LV2862YDDC

C34 10uF 63V

C35 10uF 63V

R46 100k

C48 0.1uF

C33 0.1uF

L1 SRR1208-470YL 2A 47uH

D6 SK220A 200V 2A

+24V

C36 1uF

C37 1uF

R44 300k

R45 10k

Vout = 0.765V (1+ (R1 / R2))

External LED status light
DC24V input
0.02A Red | 0.02A Yellow | 0.01A Green
4 wires
#1 brown wire = Red LED
#2 white wire = Yellow LED
#3 blue wire = Green LED
#4 black wire = Common line DC24V

J13 43650-0415 Micro-Fit

BROWN_WIRE
WHITE_WIRE
BLUE_WIRE
BLACK_WIRE

F10 SF-1206SA250M-2

LED_RED — Q4 2N7002 — R43 10k

LED_YELLOW — Q3 2N7002 — R39 10k

LED_GREEN — Q2 2N7002 — R32 10k

External Status Light

Mounting Holes

H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad

FID1 Fiducial FID2 Fiducial FID3 Fiducial

Mark Funderburk
Precesion Sustainable Agriculture

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